

Ethanol Blending in Gasoline – Taiwan

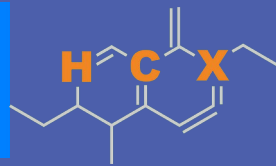
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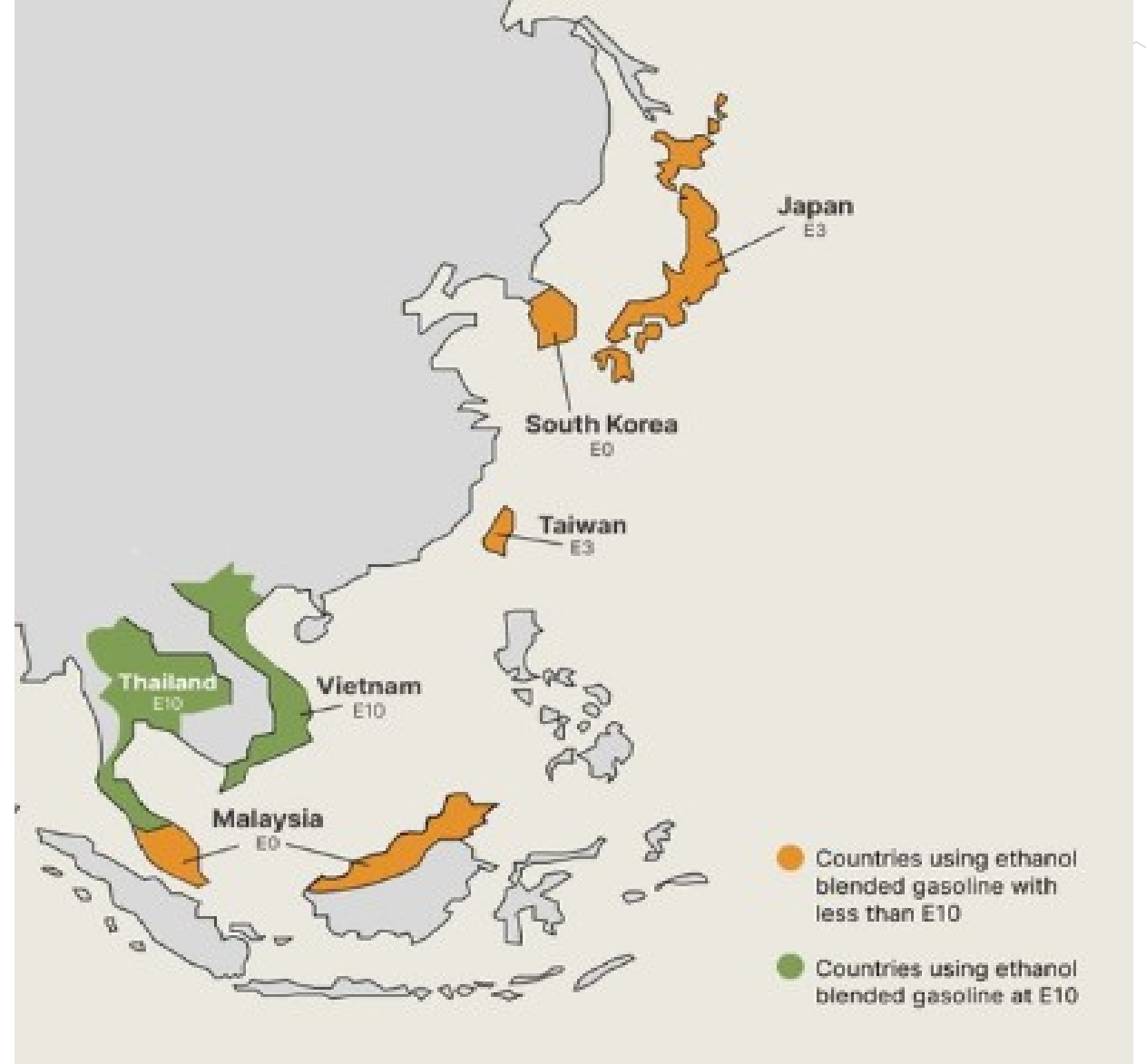


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Ethanol Blending in Asia

There are important fuel quality and environmental impact of vehicle emission challenges in these regions.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Asia to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

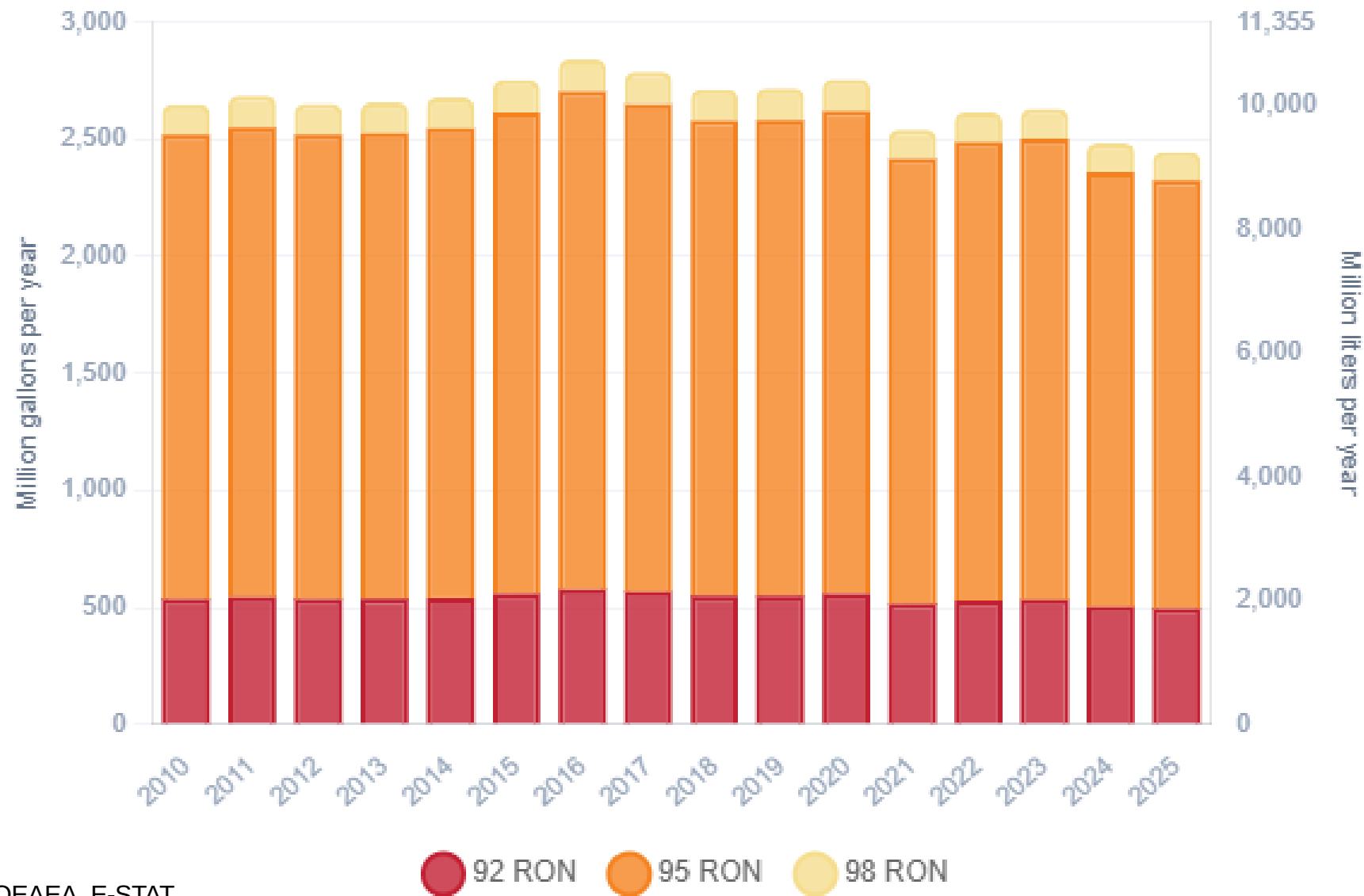
Ethanol Gasoline Blending in Taiwan



In 2025, gasoline consumption in Taiwan was 9,262 million liters (2,447 million gallons) and growth rate was -0.5%. Gasoline production and net imports were 13,152 and -4,064 million liters (3,475 and -1074 million gallons) correspondingly. Taiwan offers three main grades: RON 92, RON 95 and RON 98) with an average octane of 94.5 RON.

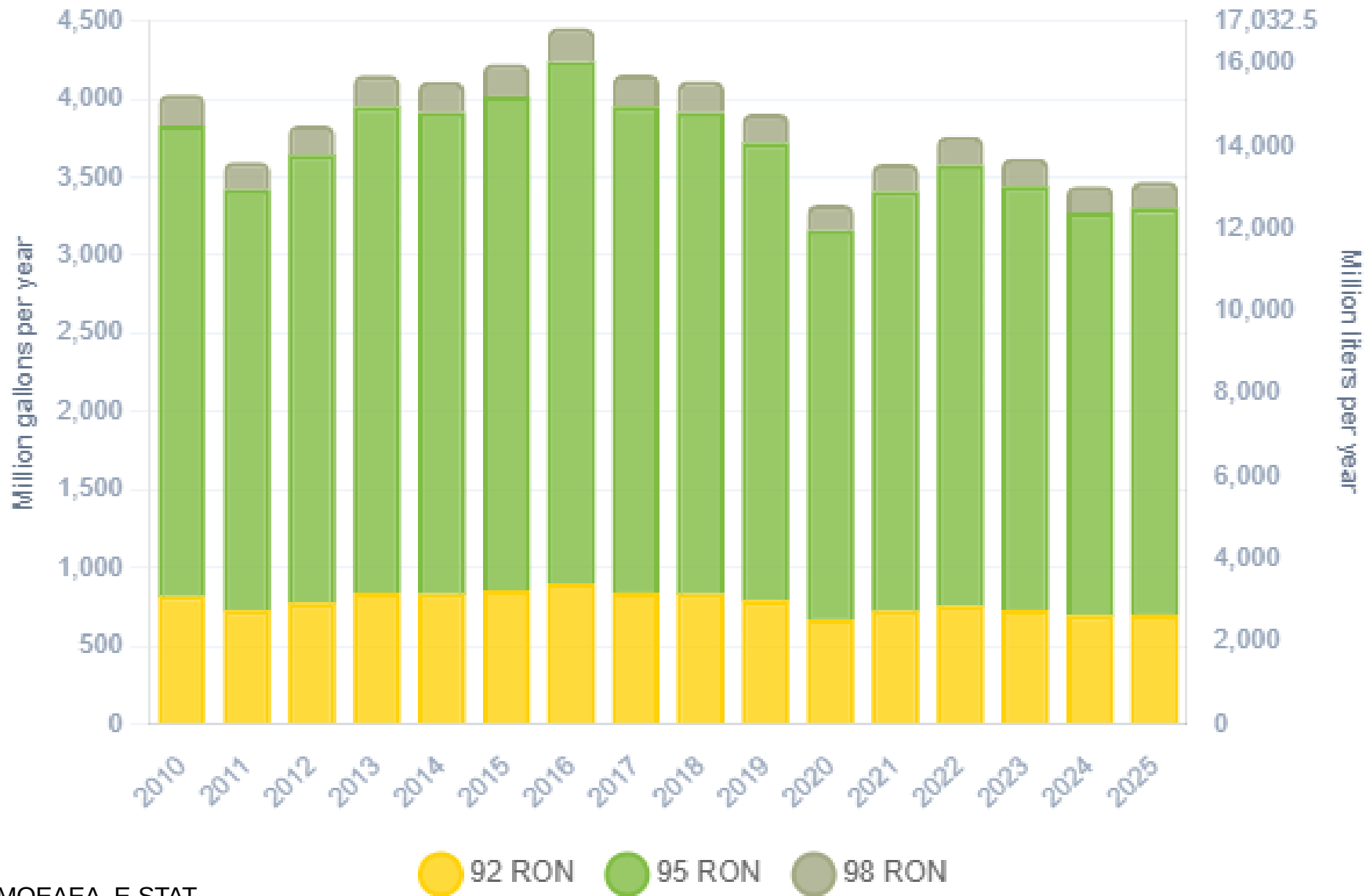
Taiwan has policies to promote biofuels in transportation to reduce imported fuel dependency and lower carbon emissions. While there was a limited trial of E3 gasohol in the mid-2000s, the lack of domestic production and insufficient incentives led to its discontinuation. Only a few gas stations offer it. Ethanol is supplied by imports and corresponds to an average content of less than 0.1%. CPC's 2026 budget indicates that the company plans to procure 96,000 liters of ethanol for E3 blends. Future bioethanol use depends on developing profitable cellulosic ethanol production from non-food waste like rice straw and food waste, rather than imported feedstocks.

Gasoline Demand in Taiwan



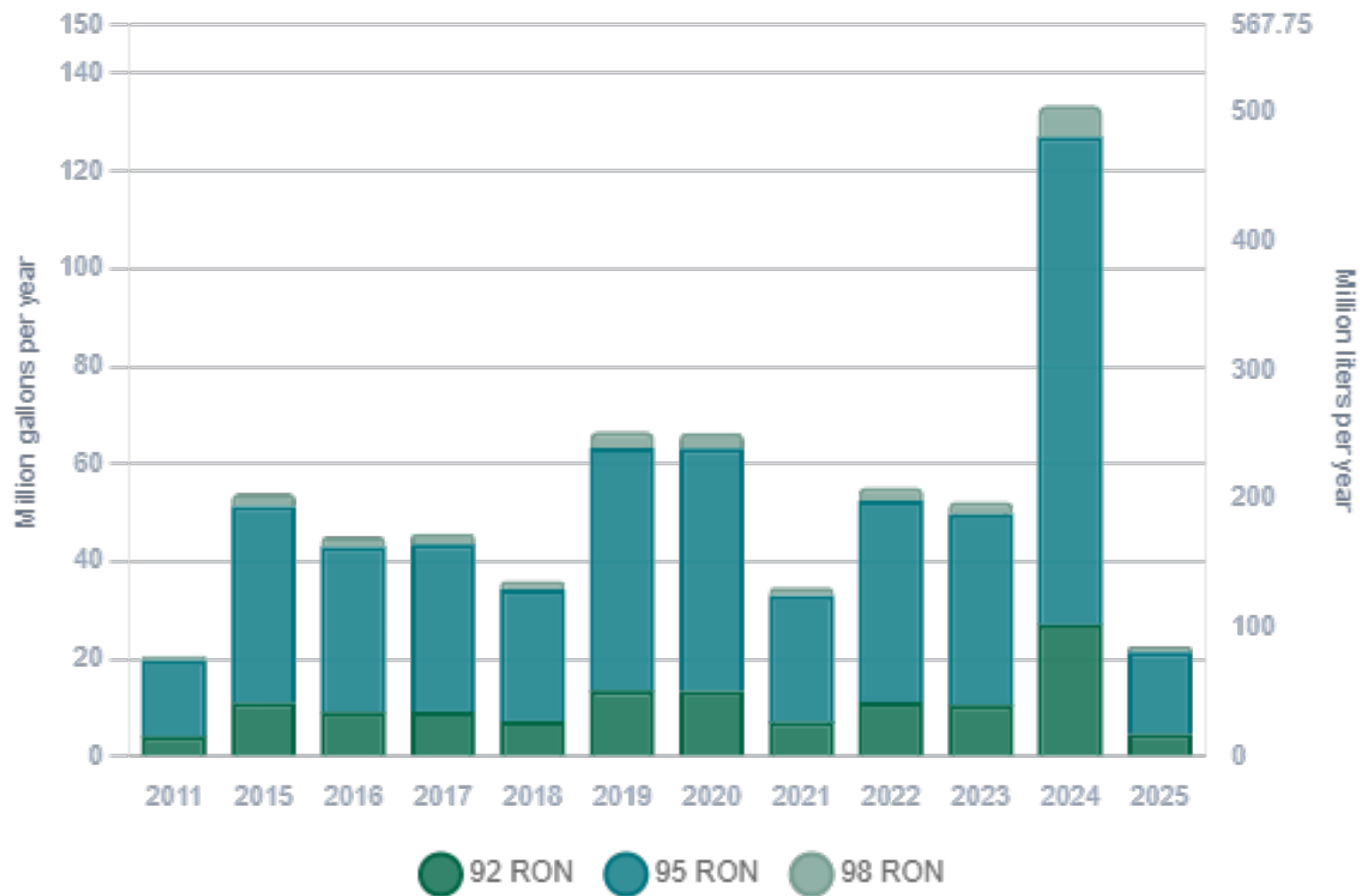
Source: MOEAEA, E-STAT

Gasoline Production in Taiwan



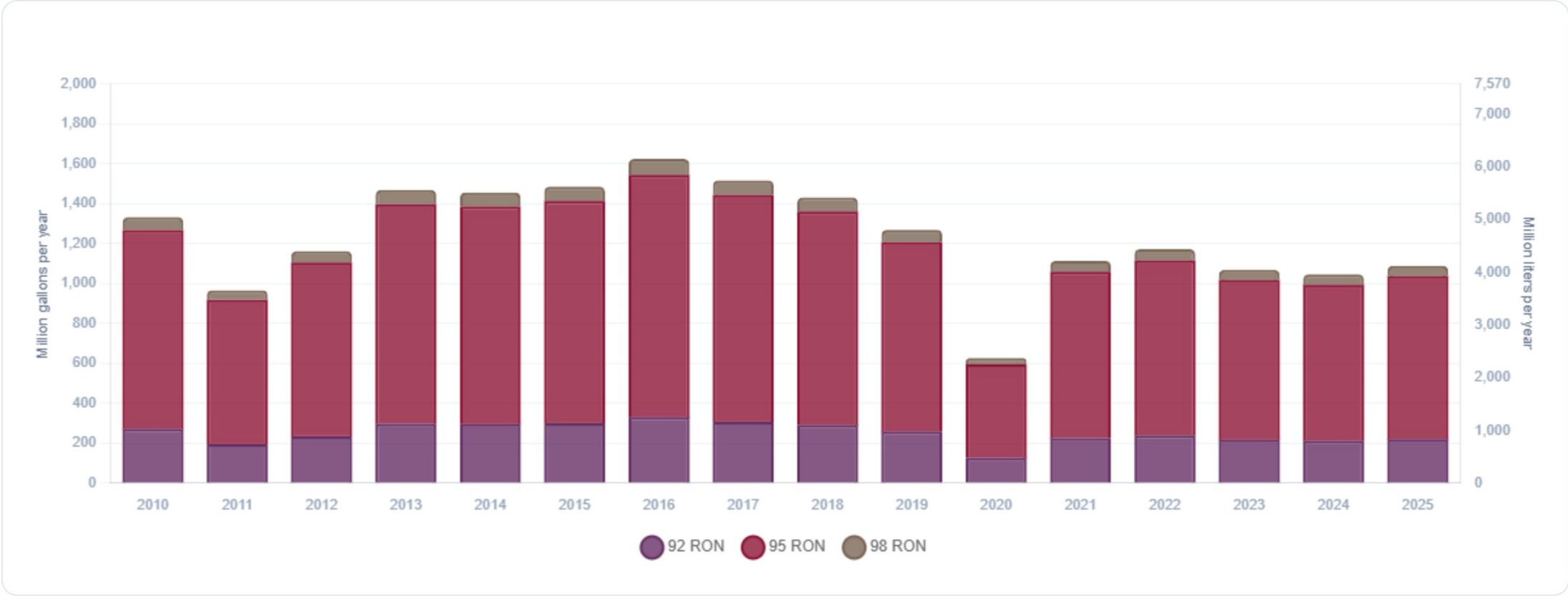
Source: MOEAEA, E-STAT

Gasoline Imports in Taiwan



Source: MOEAEA, E-STAT

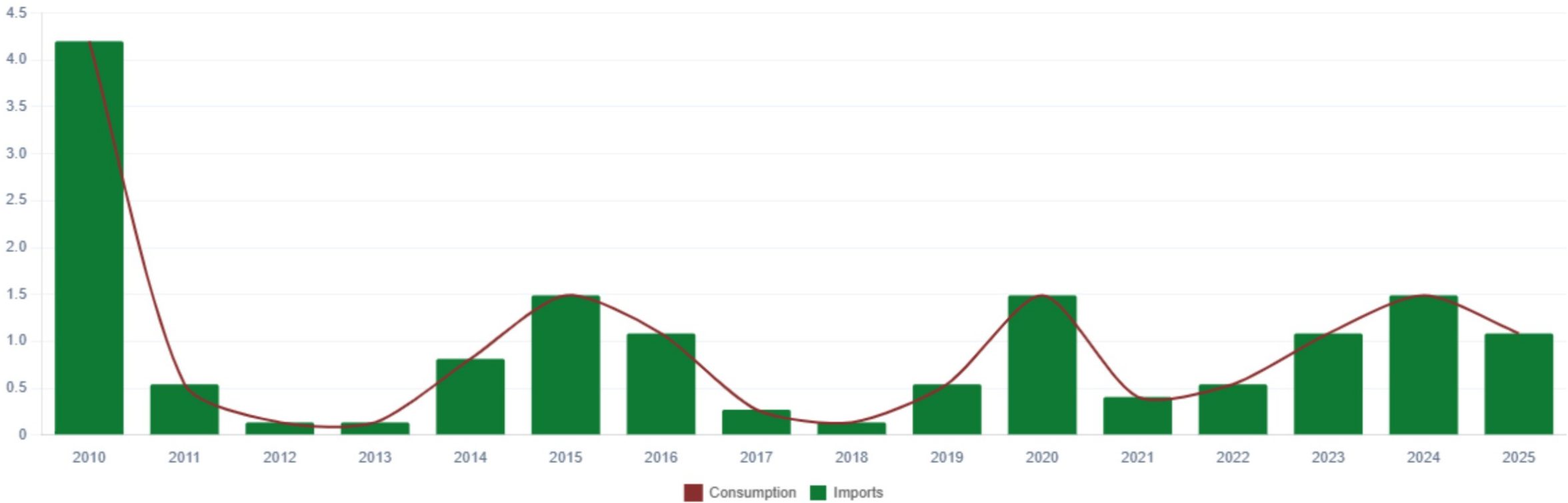
Gasoline Exports in Taiwan



Source: MOEAEA, E-STAT

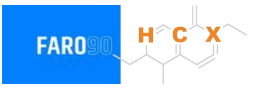
Ethanol Balance in Taiwan

Currently E3 blends are used in a small proportion of gas stations with a subsidy. There is no local production. Ethanol is supplied by imports. Country level ethanol content is 0.1%.



Source: MOEAEA, E-STAT

Gasoline Quality in Taiwan

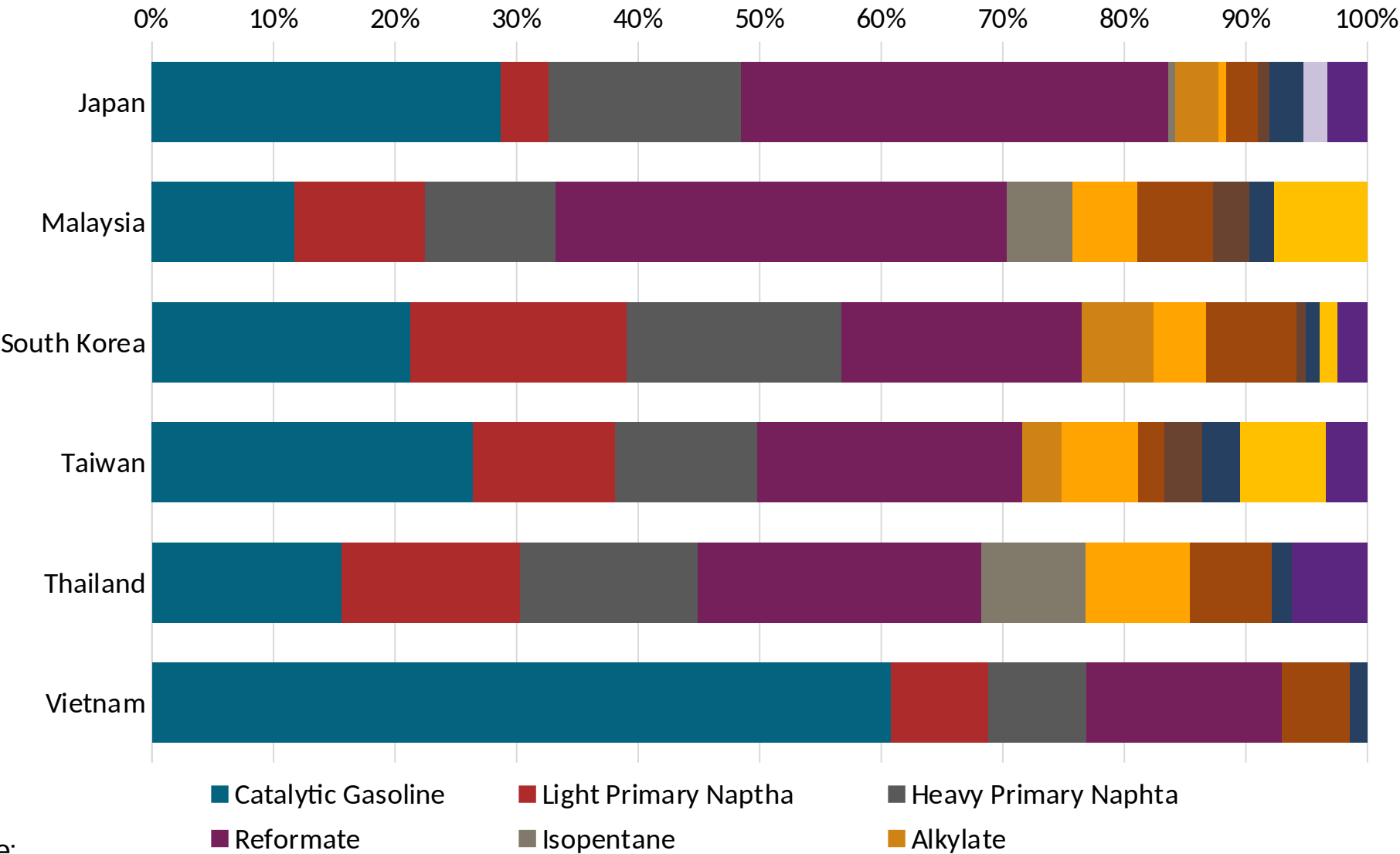


Source: MOEAEA, MOENV, Formosa Petrochemical

Gasoline Component Blending in Asia

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source:
Faro90

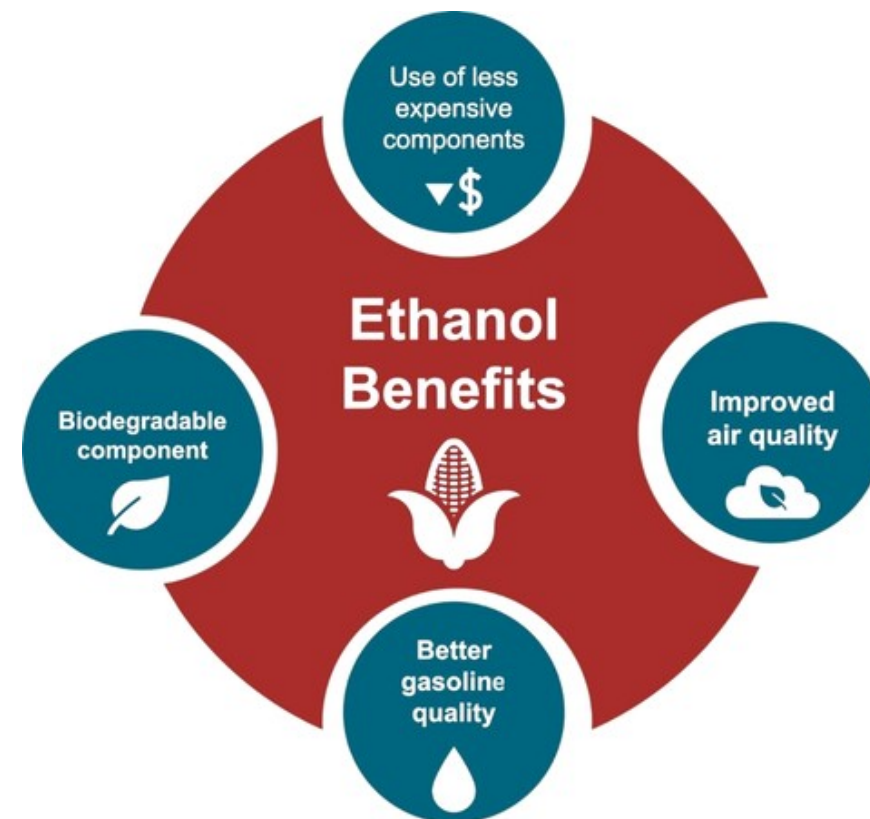
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

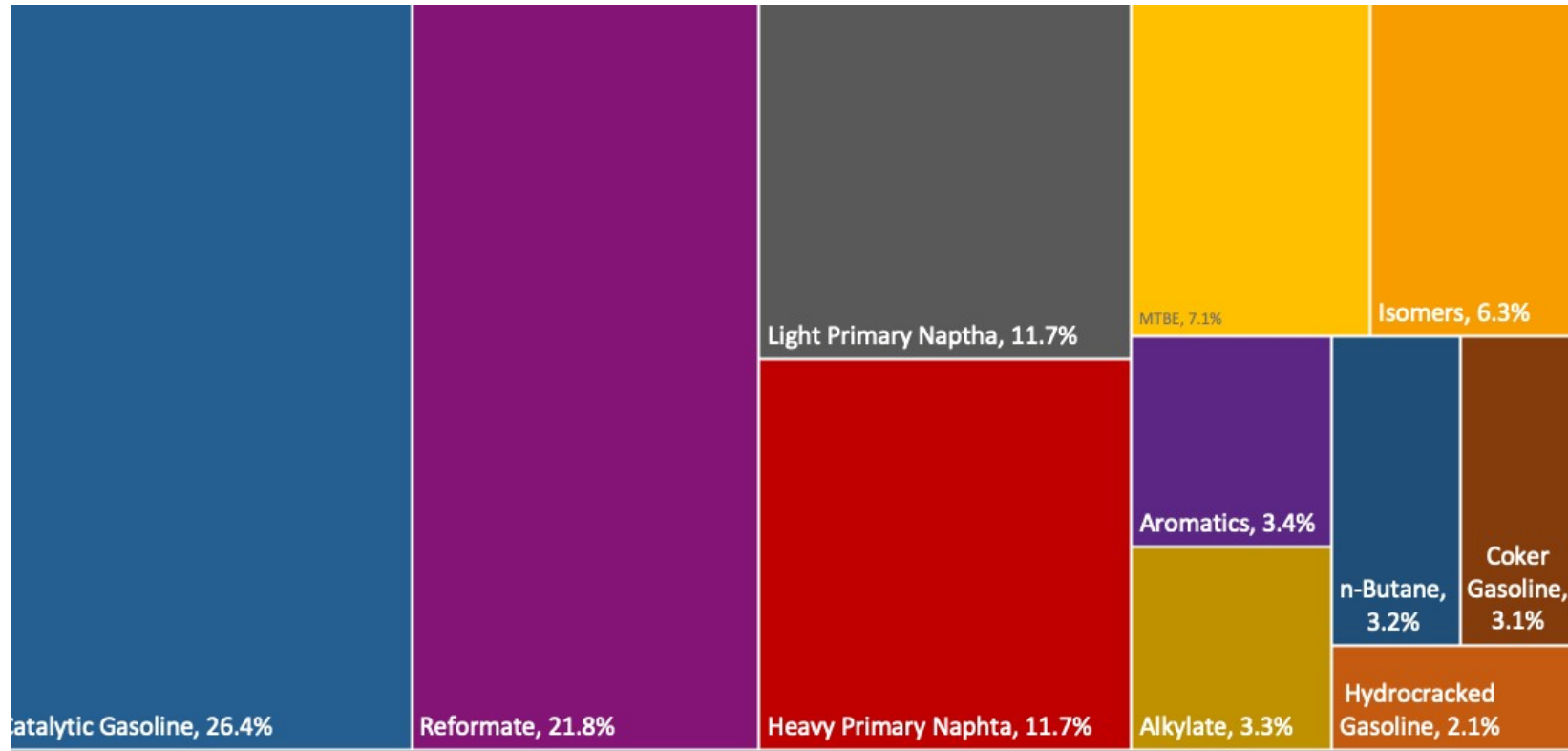
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

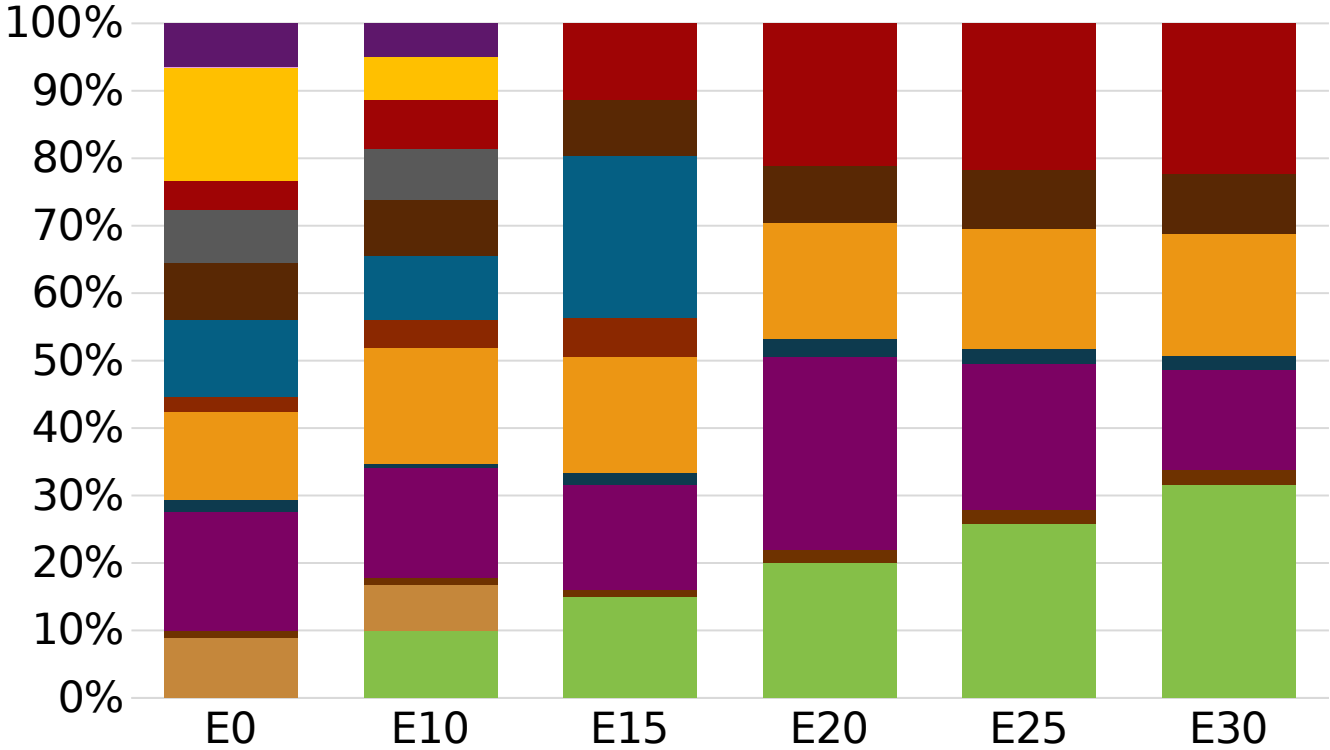
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Taiwan Available Blending Components



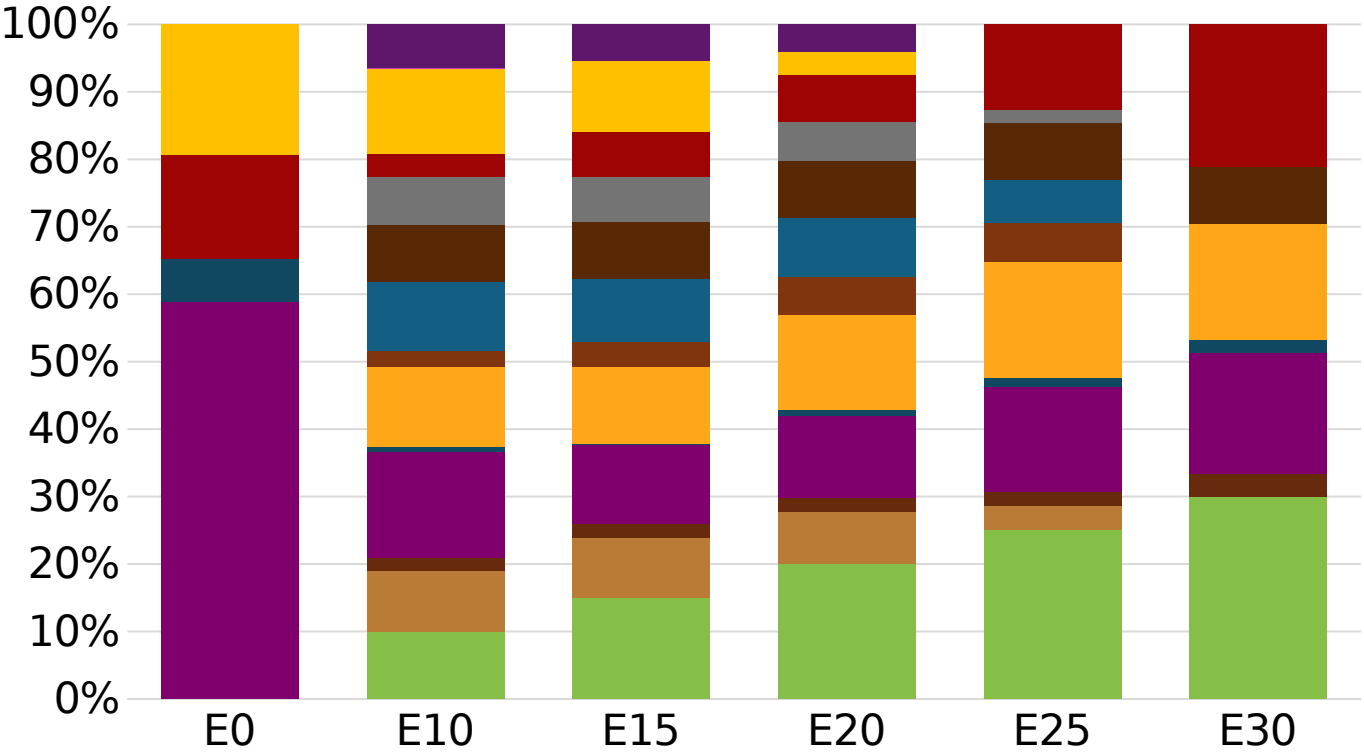
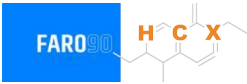
Taiwan – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.8	95.0	95.0
Price (US\$/gal)	\$2.17	\$2.09	\$2.01	\$1.94	\$1.85	\$1.78

Taiwan - Gasoline 98 - Constant Octane

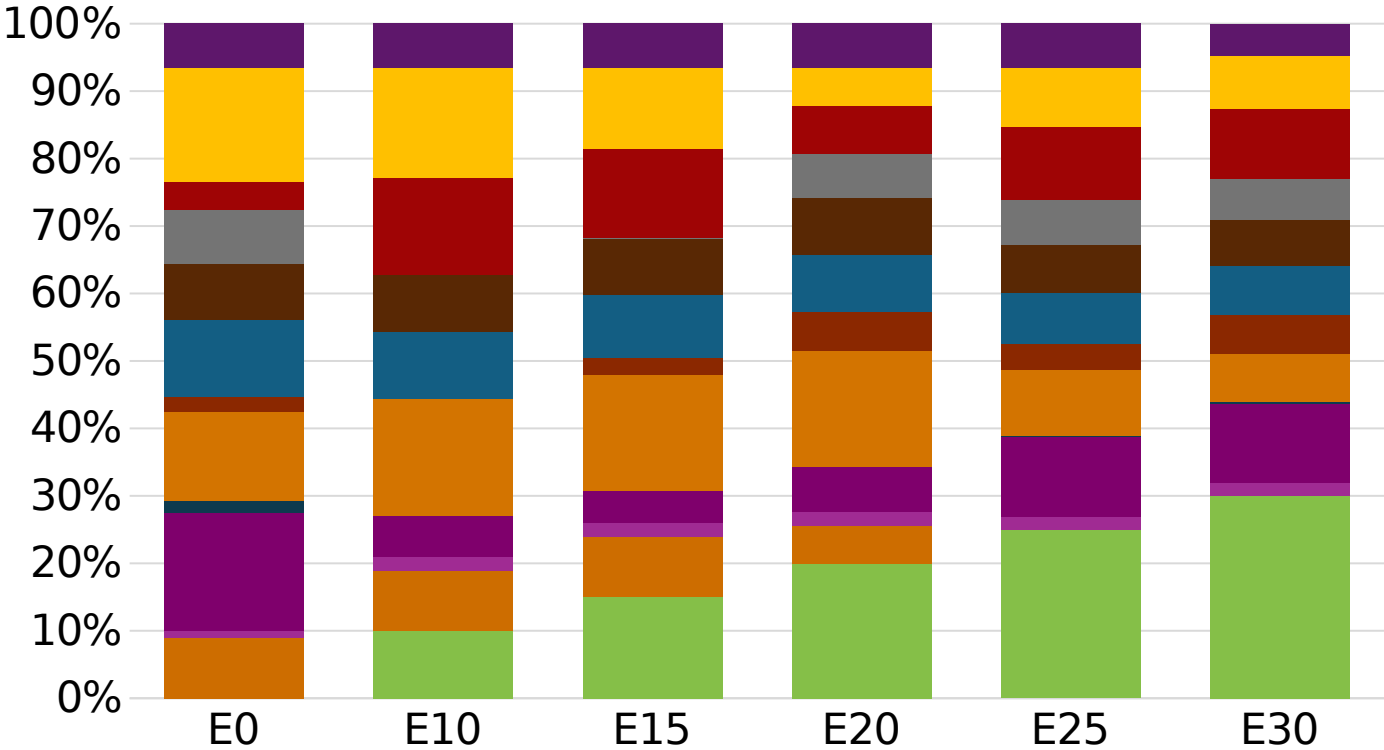


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.6
Price (US\$/gal)	\$2.23	\$2.21	\$2.13	\$2.09	\$2.00	\$1.90

Source:
Faro90

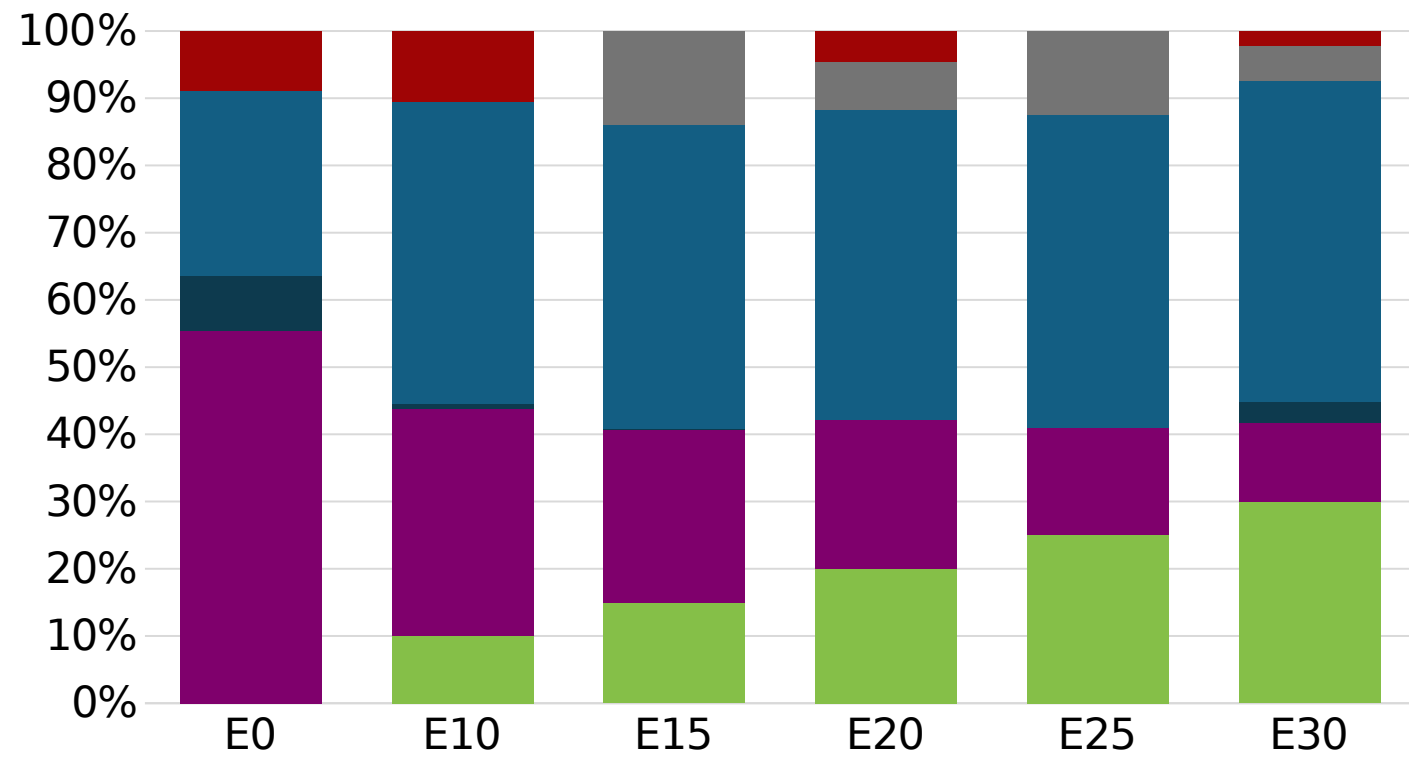
Taiwan – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	95.0	96.4	97.4	98.1	100.6	102.0
Price (US\$/gal)	\$2.17	\$2.17	\$2.17	\$2.17	\$2.17	\$2.17

Taiwan – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.8	99.9	101.3	102.6	103.9
Price (US\$/gal)	\$2.23	\$2.23	\$2.23	\$2.23	\$2.23	\$2.23

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

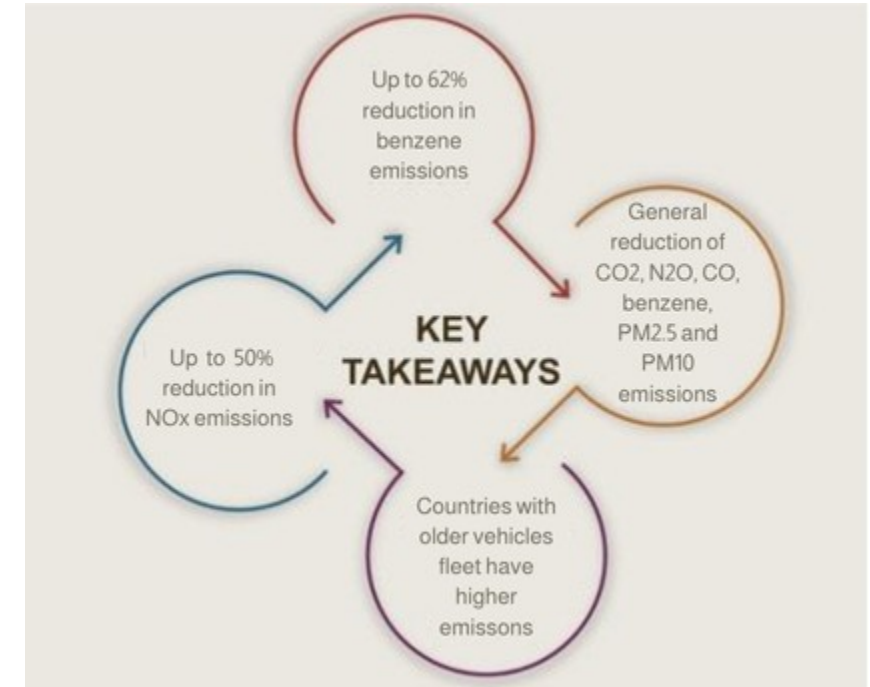
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

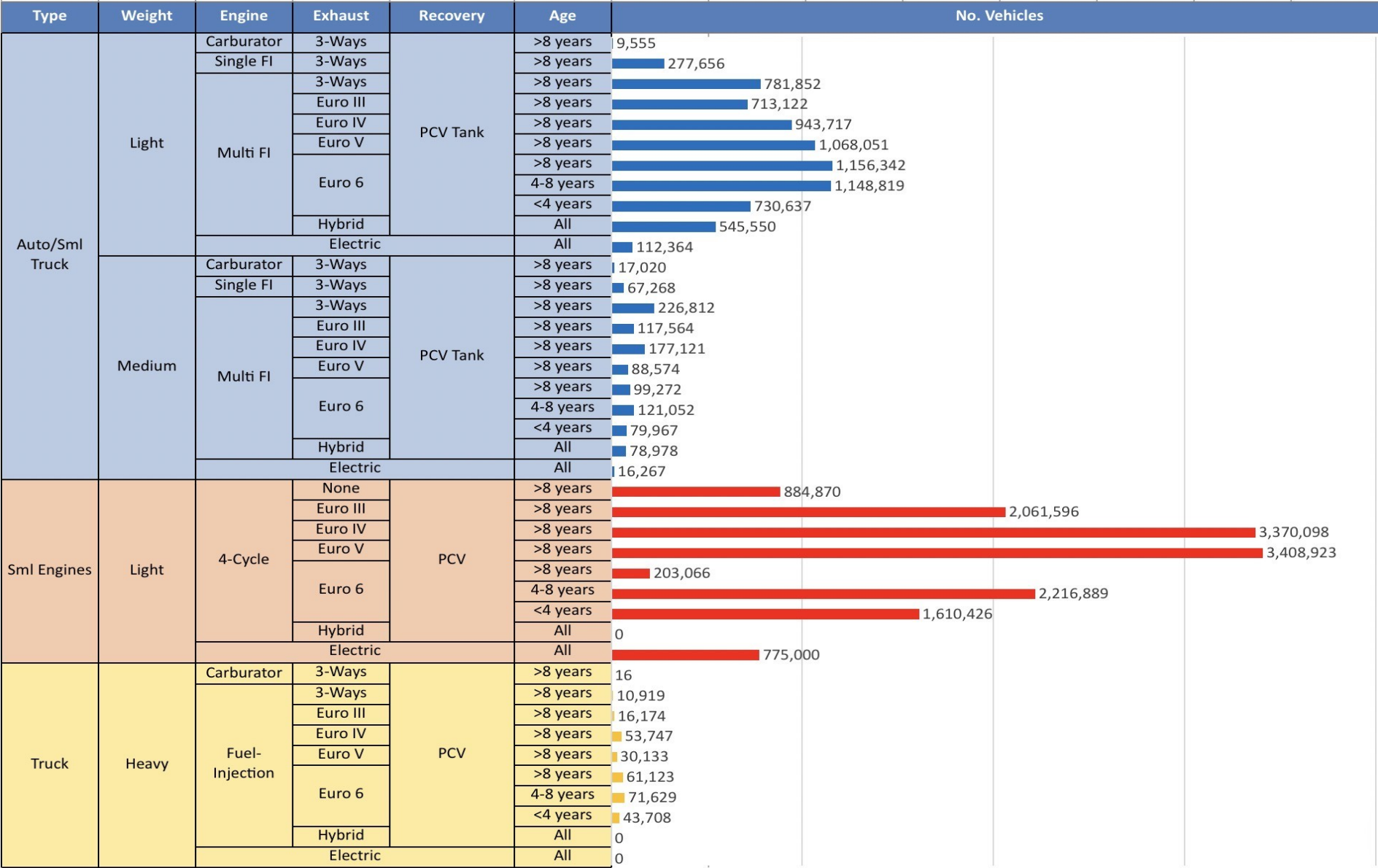
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Taiwan

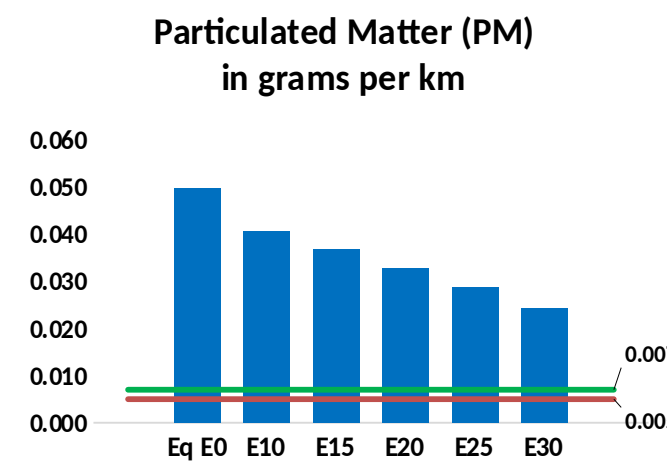
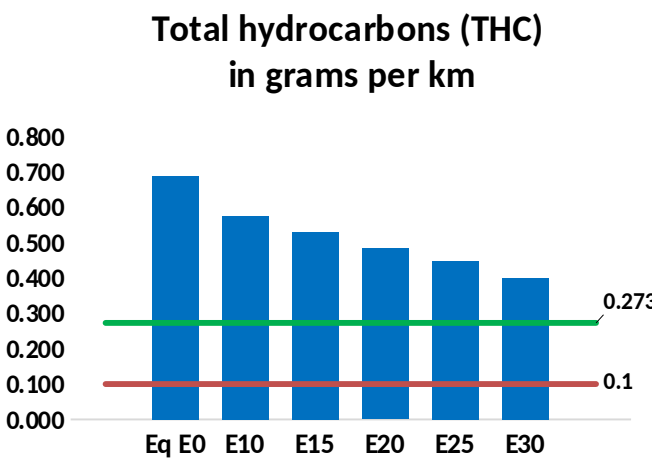
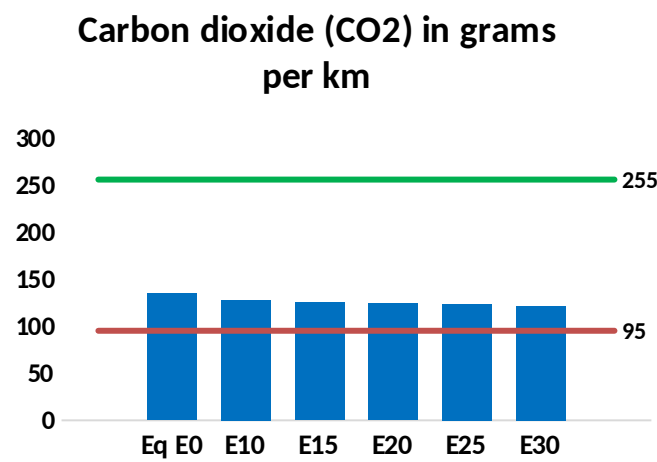
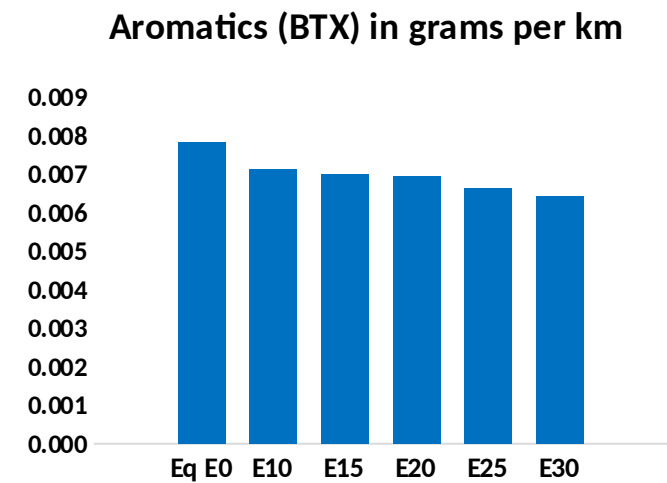
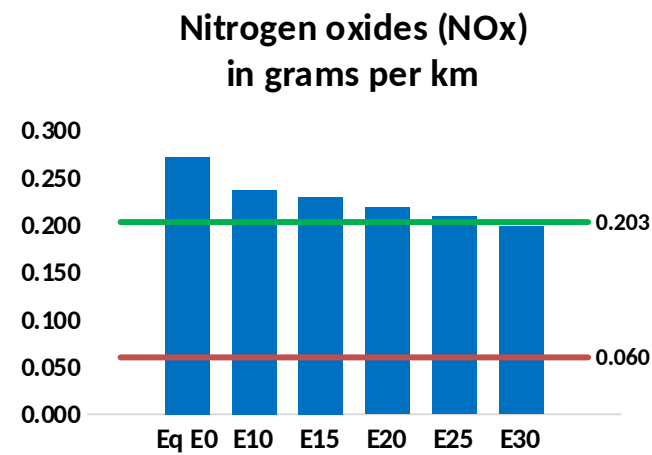
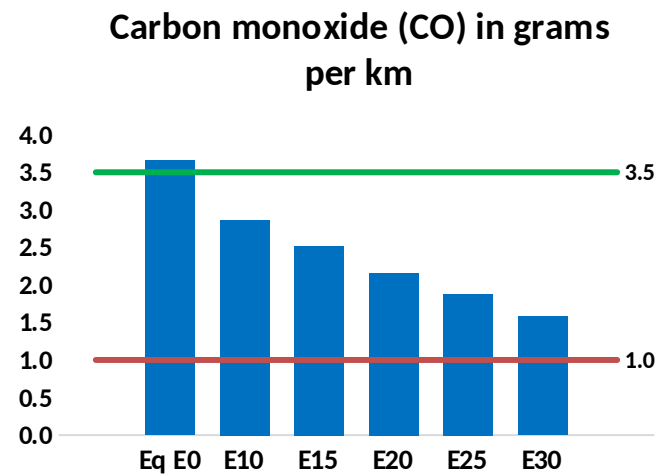
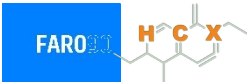


Vehicle Fleet: 23,395,876
Gasoline Vehicle Fleet: 20,722,999

Source: DATA-GOV.tw, CIECData

Taiwan – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Taiwan – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,784,176	1,590,911	1,397,645	1,228,325	1,052,895	916,478	772,360	-22%	-41%
CO2	65,534,716	63,849,538	62,257,981	61,006,268	60,386,871	60,108,716	59,000,740	-5%	-8%
VOC	335,531	308,344	281,156	258,605	235,786	218,202	194,870	-16%	-30%
NOx	132,659	122,489	115,583	112,103	106,455	102,047	96,844	-13%	-20%
SOx	716	662	608	562	518	471	421	-15%	-28%
NH3	27,896	27,620	27,344	27,474	27,783	28,000	28,181	-2%	0%
N2O	888	883	879	884	902	912	922	-1%	2%
CH4	88,450	88,450	88,450	89,777	91,722	93,403	94,995	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,196	2,040	1,885	1,760	1,635	1,539	1,410	-14%	-26%
Acetaldehyde	7,016	8,052	11,815	17,992	24,512	28,010	33,097	68%	249%
Formaldehyde	28,328	27,541	32,105	37,698	39,494	43,691	47,563	13%	39%
Benzene	3,818	3,669	3,476	3,421	3,393	3,245	3,140	-9%	-11%
PM2.5	4,571	4,156	3,740	3,374	3,017	2,633	2,232	-18%	-34%
PM10	19,701	17,910	16,119	14,543	13,001	11,348	9,618	-18%	-34%

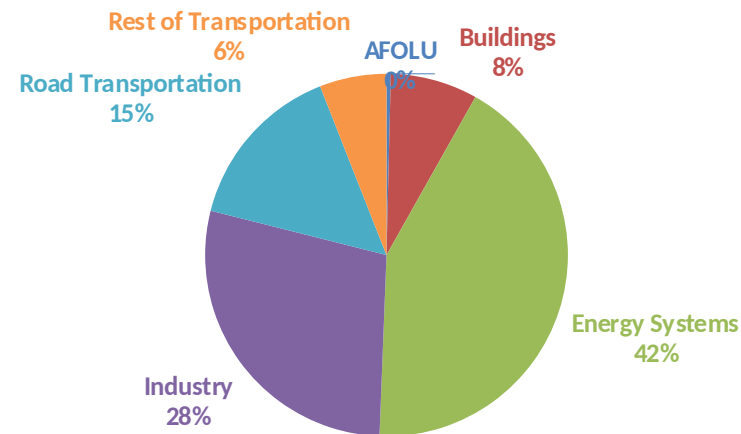
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

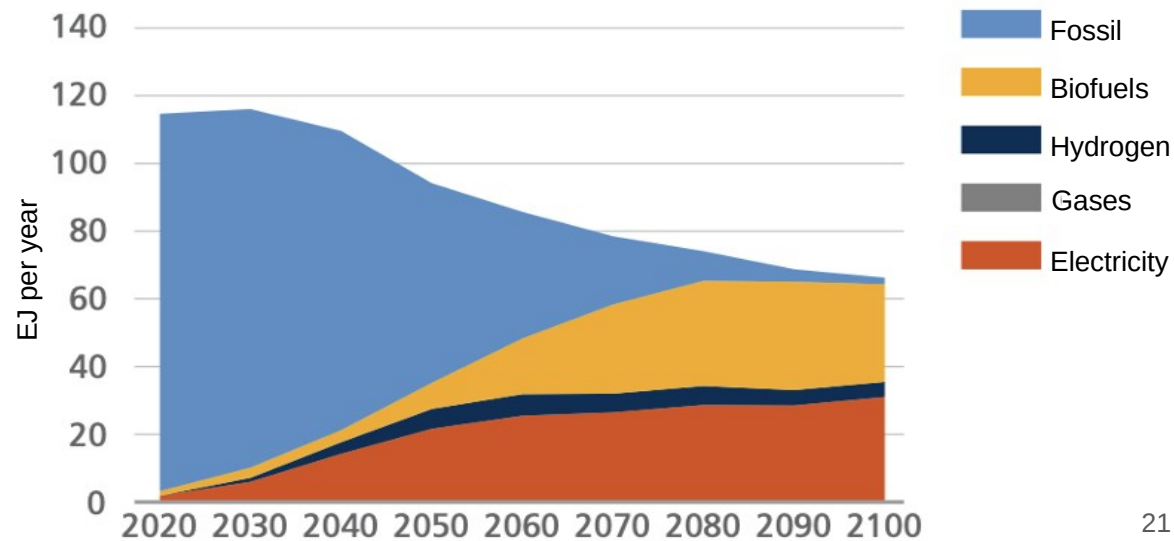
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

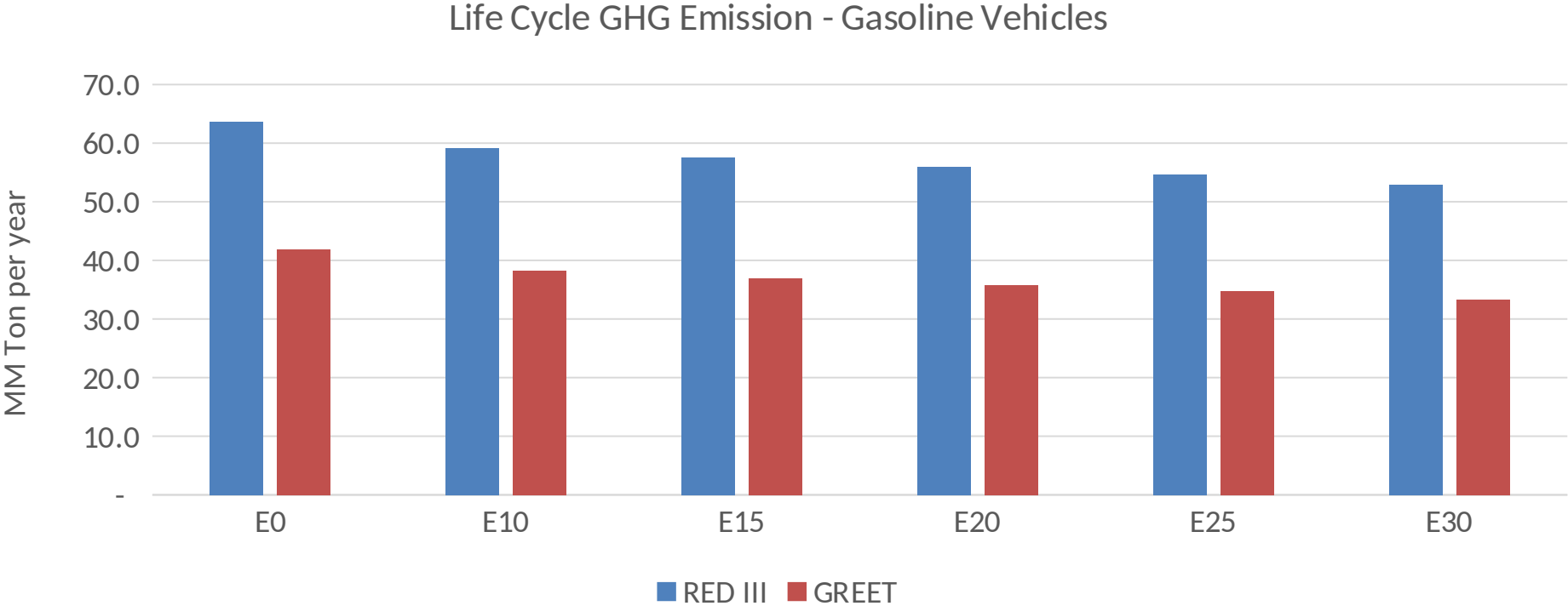
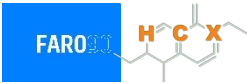
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Taiwan – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	291.7
Target @ 2035 (MMT/year)	204.2
Delta	87.5

Source: Greet, RED II/III, climateactiontracker.org, F90